



examGenome CheatSheet on Algebra

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1	What is data in statistics? Answer: Data is information collected about things, like numbers of pets, favorite colors, or test scores. Explanation: Data can be numbers (like ages) or words (like names of animals). We organize it to find patterns or make decisions.
2	What are the three main types of averages in statistics? Answer: Mean, median, and mode. Explanation: Mean = total $\div$ number of items. Median = middle value when ordered. Mode = most common value. Each shows a 'typical' result in a different way.
3	What is the range of a data set? Answer: The range is the difference between the largest and smallest values. Explanation: It tells you how spread out the data is. A small range means numbers are close together; a large range means they are far apart.
4	How do we display data clearly? Answer: Using charts and graphs like bar graphs, pictograms, line plots, and pie charts. Explanation: These pictures make it easier to see patterns, compare groups, and understand information at a glance.
5	What is probability? Answer: Probability is how likely something is to happen—like flipping a coin or rolling a die. Explanation: It's measured as a number from 0 (impossible) to 1 (certain), or as a fraction, decimal, or percentage.
6	What does a probability of 0 or 1 mean? Answer: 0 means it will never happen. 1 means it will definitely happen. Explanation: For example, the probability of rolling a 7 on a regular die is 0. The probability of the sun rising tomorrow is 1.
7	How do you write the probability of an event? Answer: As: (Number of favorable outcomes) $\div$ (Total number of possible outcomes). Explanation: Example: A coin has 2 sides. Probability of heads = 1 (favorable) $\div$ 2 (total) = $1/2$.
8	Can probability be shown as a fraction, decimal, or percent? Answer: Yes! All three are used—like $1/2$, 0.5, or 50%—to show how likely something is. Explanation: They all mean the same thing, just written differently. You can convert between them.
9	What is a fair game or experiment in probability? Answer: A fair game is one where all outcomes are equally likely—like a fair coin or a fair die. Explanation: In a fair coin toss, heads and tails each have a 50% chance. No side is more likely to win.

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10	Do statistics and probability use units? Answer: Statistics often uses the same units as the data (like cm or dollars). Probability has no units—it's just a number between 0 and 1. Explanation: For example, average height might be 140 cm (with units), but the probability of rain is 0.7 (no units).

